The Exact Minimal Time for One-End Observation and Control of a Finite String

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Abstract

We determine the sharp one-end observation and boundary-control time for the one-dimensional Dirichlet wave equation. At that exact time the endpoint normal derivative satisfies a Parseval identity with constant $4/c^3$, giving observability including the critical equality.

1 Frozen string and its full revival

Let $L, c > 0$ and consider

$$u_{tt} - c^2 u_{xx} = 0, \quad 0 < x < L, \qquad u(0,t) = u(L,t) = 0. \tag{1}$$

The adjoint data lie in $H_0^1(0,L) \times L^2(0,L)$, with conserved energy

$$E(t) = \frac{1}{2} \int_0^L \left(|u_t(x,t)|^2 + c^2 |u_x(x,t)|^2 \right) dx. \tag{2}$$

The observation is $u_x(L,\cdot)$. There is no zero mode: the frequencies are $\omega_n = n\pi c/L$, $n \geq 1$.

Lemma 1 (Least full-space revival)**.** *The least positive time for which the wave group is the identity on the full energy space is*

$$T_* = \frac{2L}{c}.$$

Proof. At time T_*, every phase $e^{\pm i\omega_n T_*} = e^{\pm 2\pi i n}$ is one. Conversely, an identity time must return the $n = 1$ mode and is therefore an integer multiple of $2L/c$. In particular, L/c multiplies the nth mode by $(-1)^n$ and is not the full-space identity. □

2 Critical endpoint Parseval identity

Every finite-energy Dirichlet solution has the periodic d'Alembert form

$$u(x,t) = F(x+ct) - F(-x+ct), \qquad F(s+2L) = F(s), \quad F' \in L^2(\mathbb{R}/2L\mathbb{Z}). \tag{3}$$

Constants in F are irrelevant. Differentiating gives

$$u_x(L,t) = 2F'(L+ct), \qquad E = c^2 \int_0^{2L} |F'(s)|^2 \, ds. \tag{4}$$

The energy equality follows by adding the squares of u_t and cu_x and changing variables over the two reflected halves.

Theorem 1 (Exact critical observability)**.** *Every solution of* (1) *satisfies*

$$\int_0^{2L/c} |u_x(L,t)|^2 \, dt = \frac{4}{c^3} E(0). \tag{5}$$

Consequently one-end boundary observability holds for every $T \geq 2L/c$, including equality.

Proof. At the critical time, $s = L + ct$ traverses one complete period. Hence

$$\int_0^{2L/c} |u_x(L,t)|^2 dt = \frac{4}{c} \int_0^{2L} |F'(s)|^2 ds = \frac{4}{c^3} E(0).$$

For a longer interval, integrate the nonnegative observation over its first critical subinterval. $\square$

For comparison, if

$$u(x,t) = \sum_{n \geq 1} \left(a_n \cos \omega_n t + \frac{b_n}{\omega_n} \sin \omega_n t \right) \sin \frac{n\pi x}{L},$$

orthogonality over $[0, 2L/c]$ gives (5) mode by mode. The periodic proof is stronger for the sharpness argument below because it does not truncate the solution.

References

[1] J.-L. Lions, *Exact controllability, stabilization and perturbations for distributed systems*, SIAM Review **30**(1) (1988), 1–68, doi:10.1137/1030001.

[2] C. Bardos, G. Lebeau, and J. Rauch, *Sharp sufficient conditions for the observation, control, and stabilization of waves from the boundary*, SIAM Journal on Control and Optimization **30**(5) (1992), 1024–1065, doi:10.1137/0330055.